

TEXAS COLLEGE OF MANAGEMENT AND IT

Affiliated to Lincoln University College



Project Synopsis of

NutriFit: AI-Powered Personalized Nepali Diet & Nutrition Recommendation System

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Project Title:

NutriFit: AI-Powered Personalized Nepali Diet & Nutrition Recommendation System

Introduction / Background:

Eating healthy has become more challenging for many people in Nepal. Although there is a lot of information available on diet and nutrition, most of it is either too general or focused on Western foods that do not match our local eating habits. Because of this, many people end up following diets that don't suit their body type, lifestyle, or health conditions. Even though people want to improve their diet, they often don't know how to plan meals using Nepali foods in a healthy and balanced way.

With the growth of artificial intelligence and machine learning, it is now possible to provide smarter and more personalized health suggestions. These technologies can analyze a person's details - such as age, weight, activity level, budget, and health conditions—and automatically recommend meals that fit them well.

My project, **NutriFit**, aims to use AI to create a personalized diet recommendation system based on Nepali foods. The system will calculate the user's BMI, daily calorie needs, food preferences, and other factors, and then generate a meal plan that suits them. It will use machine learning techniques like K-Means clustering and content-based recommendation methods to give suggestions that are practical and easy to follow.

Problem Statement:

In Nepal, many individuals struggle to maintain healthy diets because there is a lack of personalized nutritional guidance that considers their body type, activity level, budget, and health conditions such as diabetes. Most available diet recommendations are generic and do not include commonly consumed Nepali foods, making them impractical and difficult to follow. As a result, people often develop poor dietary habits and fail to plan balanced meals effectively. There is a need for an AI-powered system that provides personalized and practical nutrition recommendations using locally available Nepali foods.

The specific problems identified are:

- Lack of platforms offering Nepali-food-based meal planning and culturally relevant dietary guidance.
- Difficulty planning balanced meals within a limited budget using affordable local foods.

- Overload of online nutrition information that users cannot convert into practical, personalized meal plans.

Objectives of the Project:

General Objective:

To develop an AI-powered web application that provides personalized diet and nutrition recommendations for Nepali users based on their body profile, activity level, dietary preferences, budget constraints, and health conditions.

Specific Objectives:

- To develop a user profiling system that calculates BMI and daily caloric needs, along with a nutritional database of commonly consumed Nepali foods.
- To develop K-Means clustering for user segmentation and content-based filtering to generate personalized meal plans.
- To design and deploy a responsive web application with an interactive dashboard.

Scope of the Project:

In-Scope:

- **User Profiling:** Registration, authentication, health metrics collection (age, gender, height, weight, activity level, dietary preferences, budget, diabetes status), BMI calculation, and caloric requirement estimation using Mifflin-St Jeor equation.
- **Food Database:** 50-60 Nepali foods with nutritional data (calories, protein, carbs, fats, glycemic index) and pricing information.
- **AI/ML Engine:** K-Means clustering for user segmentation and content-based filtering for meal recommendations with dietary, budget, and diabetes-aware filtering.
- **Meal Planning & Visualization:** Daily meal plan generation (breakfast, lunch, dinner) with detailed nutritional breakdown, responsive web-based user interface with interactive dashboard for displaying recommendations, and PostgreSQL database integration.
- **Deployment & Documentation:** Cloud platform deployment and complete project documentation.

Additional Features (if time permits):

- Interactive data visualization charts for nutritional analysis
- Admin panel for food database management
- Meal plan regeneration functionality

Out of Scope:

- Real-time health monitoring through wearable devices
- Advanced conversational AI chatbot
- Image-based food recognition and calorie tracking
- Extensive meal history tracking and long-term progress analytics
- User meal rating system

Significance of the Project:**For Individual Users:**

- Makes personalized nutrition guidance easily available without the need for costly nutritionist consultations.
- Helps users choose suitable foods based on their health condition, lifestyle, dietary preferences, and budget.

For Society:

- Encourages healthy eating habits using affordable and locally available Nepali foods.
- Promotes preventive healthcare by increasing awareness of balanced nutrition among the general population.

Academic Relevance:

- Demonstrates the practical use of artificial intelligence and machine learning techniques in a real-world health application.
- Applies methods such as K-Means clustering and content-based recommendation systems in the nutrition domain.

Literature Review:**Study 1: MyFitnessPal - Calorie Tracking Application**

MyFitnessPal is one of the most popular nutrition tracking applications with over 200 million users worldwide. It provides calorie counting, macro tracking, and a vast database of foods.

Advantages: Comprehensive food database, barcode scanning, community support

Limitations: Primarily focused on Western foods, requires manual logging of meals, lacks personalized AI-based recommendations, does not consider glycemic index for diabetic users, subscription-based premium features

Gap: Does not include Nepali foods, no AI-powered personalization based on user clustering.

Study 2: Nutrino - AI-Powered Nutrition Coach

Nutrino uses machine learning to provide personalized meal recommendations based on user goals, preferences, and health data.

Advantages: AI-driven recommendations, considers health conditions, integrates with fitness trackers

Limitations: Generic food database without cultural specificity, expensive subscription model, requires constant data input from wearables

Gap: No support for Nepali cuisine, lacks budget-conscious recommendations, not accessible to users without wearable devices

Research Gap Identified:

Despite numerous nutrition applications and research in dietary recommendations, there exists a significant gap in:

- AI-powered systems specifically designed for Nepali/South Asian cuisines
- Budget-conscious meal planning for economically diverse populations
- Integration of glycemic index awareness with traditional food databases
- Accessible platforms that don't require expensive wearables or subscriptions

My project '**NutriFit**' fills these gaps by combining K-Means clustering for user segmentation with content-based filtering for meal recommendations, specifically tailored to Nepali foods, budgets, and health conditions including diabetes.

Methodology:

The '**Iterative Development Model**' is chosen for this project as it allows incremental development and refinement of system modules, while enabling documentation to be prepared alongside each phase. This approach is suitable for AI-powered applications and aligns with the academic timeline, allowing supervisor feedback after each iteration.

Project Iterations:

- Iteration 1: Requirement analysis, environment setup, authentication module, and food database creation; initial documentation.

- Iteration 2: BMI and caloric calculation, K-Means clustering, recommendation engine, REST API development; update documentation
- Iteration 3: Frontend dashboard development, filtering integration, full system integration; document integration and UI.
- Iteration 4: Testing, deployment and final documentation.

Tools and Technologies:

Programming Languages:

- Python (Backend, Machine Learning)
- JavaScript (Frontend)
- SQL (Database queries)

Framework:

- React.js (Frontend)
- Django (Backend)
- Django REST Framework: For building RESTful APIs to connect frontend and backend
- Tailwind CSS

Database:

- PostgreSQL

Development Tools:

- VS Code: Integrated Development Environment
- Git & GitHub: Version control and code repository

System Requirements:

Hardware Requirements:

- **Processor:** Intel Core i5 or equivalent
- **RAM:** 8 GB
- **Storage:** 256 GB SSD

Software Requirements:

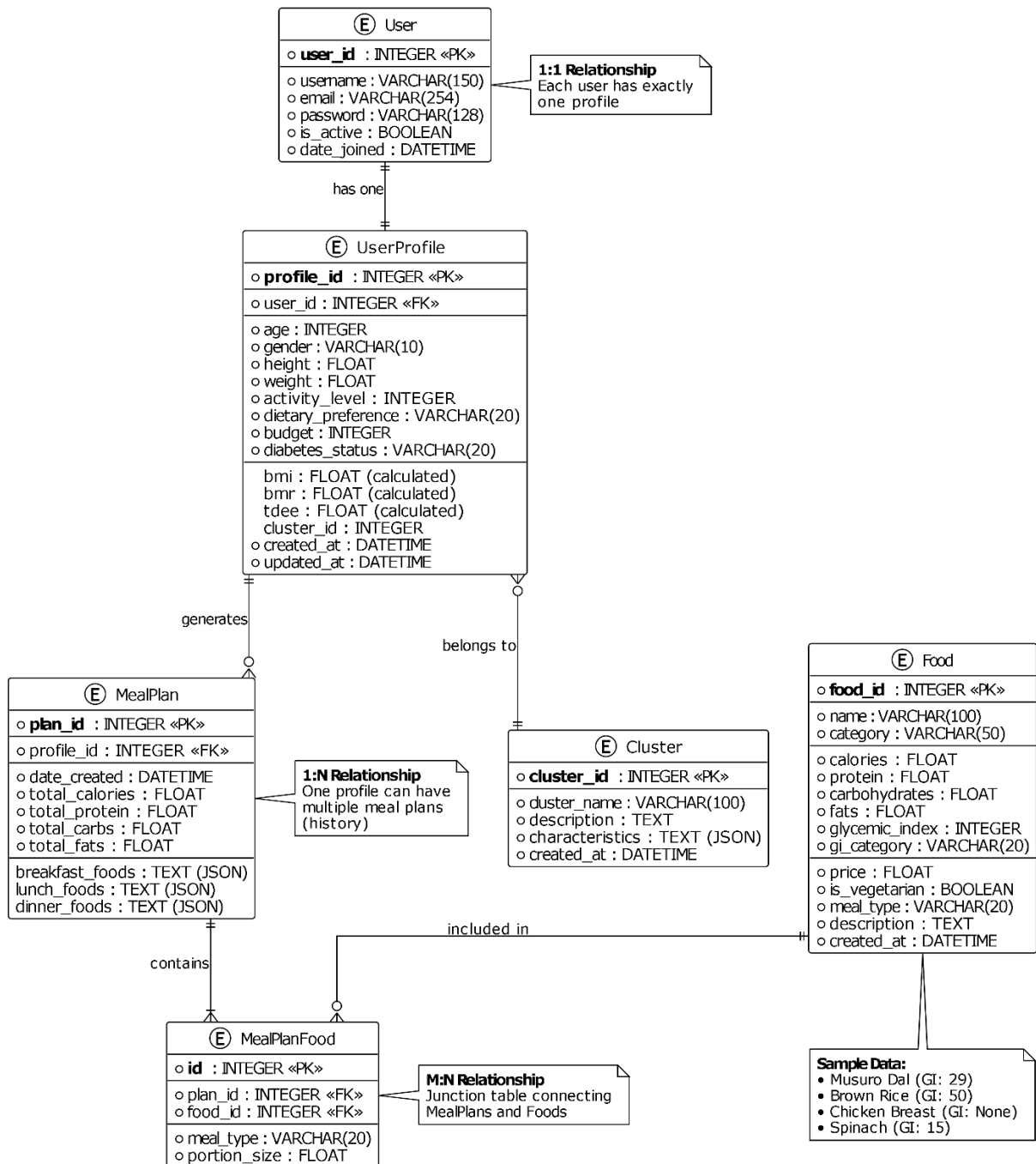
- **Operating System:** Windows 10/11, macOS, or Linux
- **IDE Tools:** Visual Studio Code
- **Database Server:** PostgreSQL

- **Browsers:** Chrome (latest version browser)

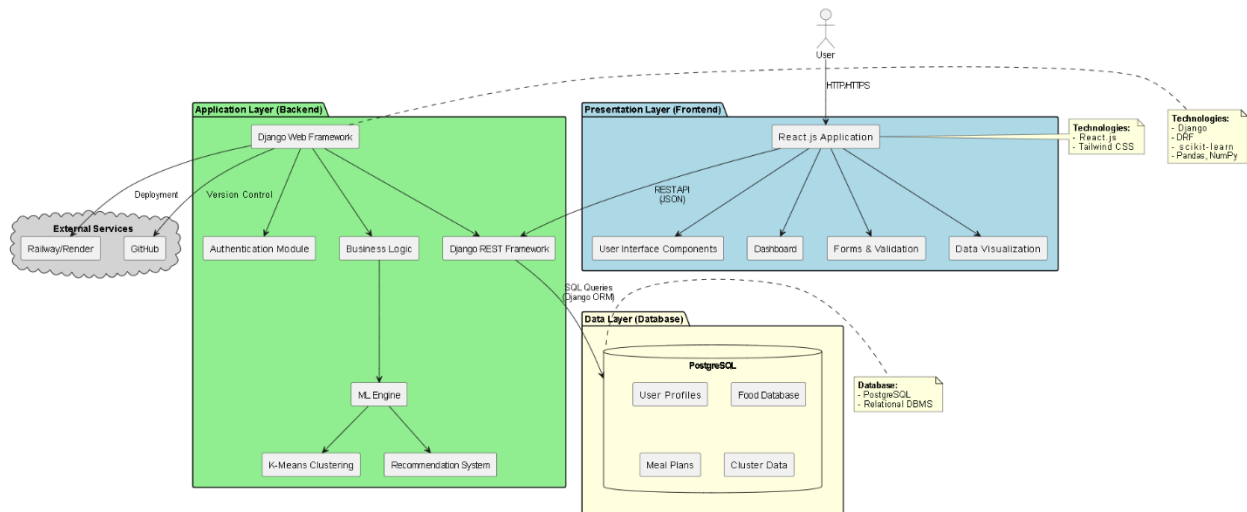
System Architecture:

ER Diagram:

Entity Relationship Diagram (ERD) NutriFit: AI-Powered Personalized Nepali Diet & Nutrition Recommendation System



Tree-Tier Architecture Diagram:



MODULES:

- **Module 1: User Authentication & Profile Management**
 - User registration, login, profile creation with health metrics, BMI calculation
- **Module 2: Food Database Management**
 - 50-60 Nepali foods with nutritional values, glycemic index, pricing, admin panel
- **Module 3: AI-Powered Recommendation Engine**
 - K-Means clustering, caloric calculation (Mifflin-St Jeor), content-based filtering, multi-constraint filtering (diet/budget/diabetes), meal plan generation
- **Module 4: User Dashboard**
 - Meal plan display, nutritional breakdown
- **Module 5: Responsive Web Interface**
 - Mobile-friendly React components with Tailwind CSS

Expected Output:

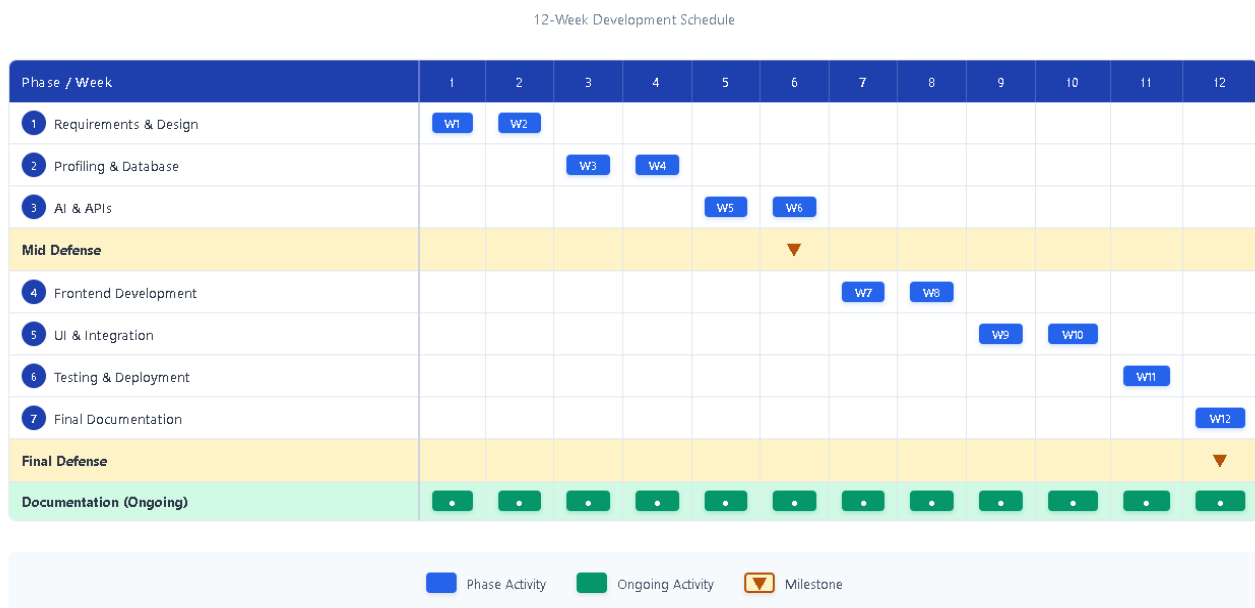
- Fully functional web application deployed on cloud platform
- Source code on GitHub with proper commit history
- PostgreSQL database with 50-60 Nepali foods
- Trained K-Means clustering model
- Complete documentation

Project Timeline / Gantt Chart:

Project Timeline:

Phase	Weeks	Activities
Phase 1	Weeks 1–2	Requirements analysis, system design, environment setup (Django, React, PostgreSQL), authentication module
Phase 2	Weeks 3–4	User profiling, BMI and calorie calculation, food database creation (50–60 Nepali foods)
Phase 3	Weeks 5–6	K-Means clustering, recommendation engine development, REST API implementation, Mid Defense
Phase 4	Weeks 7–8	React dashboard development, meal visualization, frontend-backend integration
Phase 5	Weeks 9–10	UI/UX enhancement, data visualization, responsive design, complete system integration
Phase 6	Week 11	Testing (unit, integration, user acceptance testing), deployment, bug fixing
Phase 7	Week 12	Final documentation, final report preparation, Final Defense

Gantt Chart:



Limitations:

Technical Limitations:

- Food database limited to 50-60 common foods initially.
- Daily meal plans are generated; weekly or monthly variety optimization is not included.
- No real-time health monitoring or wearable integration.
- English-only interface (Nepali language support planned for future)

- Requires internet connectivity (no offline mode)

Functional Limitations:

- Does not replace professional medical consultation.
- Does not account for food allergies or precise individualized portions.

Future Enhancements:

- Expand food database to 150-200 items with regional specialties
- Nepali language support
- Mobile application (Android/iOS)
- Weekly meal planning
- Recipe details with cooking instructions
- Meal regeneration functionality
- Integrate wearable devices to incorporate activity data for dynamic calorie adjustments.

Conclusion:

NutriFit: AI-Powered Personalized Nepali Diet & Nutrition Recommendation System addresses the critical need for accessible, culturally appropriate nutritional guidance in Nepal by combining AI/ML algorithms with a local food database. The system's ability to personalize recommendations based on individual profiles and health conditions, particularly diabetes, demonstrates practical application of machine learning in healthcare. Within 12 weeks, we will deliver a functional prototype that empowers users to make informed dietary choices aligned with their health goals and cultural practices.

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